Tertiary Infotech Academy Pte Ltd
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LEARNER GUIDE

For

AWS Certified SysOps Administrator Associate Training

TGS Ref No: TGS-2024051413

Conducted by
Tertiary Infotech Academy Pte Ltd
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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
3.0	Legacy release	Legacy Learner Guide retained as source evidence.	Dr. Alfred Ang
4.0	30 July 2026	Previous 927-slide SOA-C02 courseware release.	Dr. Alfred Ang
5.0	23 August 2026	Rebuilt visual SOA-C03-aligned deck; added native charts, ImageGen visuals, twelve self-contained AWS labs scheduled at four per day with editable templates and JPEG references, detailed Learner Guide, aligned Lesson Plan and revised WA plus Case Study assessments.	Dr. Alfred Ang

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How to Use This Guide

The slide deck explains systems, relationships and operational decisions. This Learner Guide and the individual lab folders contain the detailed procedures, command examples, evidence checkpoints and acceptance criteria.

- Read the topic explanation before starting its aligned lab.
- Work only in the trainer-authorised learner AWS account and Region.
- Replace all REPLACE placeholders with learner-owned identifiers.
- Capture evidence before cleanup, then verify that chargeable resources are deleted.
- Never place access keys, passwords, account IDs or customer data in a lab submission.

Course and WSQ Alignment

Field	Approved value
Course	AWS Certified SysOps Administrator Associate Training
Course code	TGS-2024051413
TSC	Cloud Computing Application (LOG-TEM-3007-1.1)
Duration	3 days · 24 hours · 22 hours facilitated delivery + 2 hours assessment
Labs	12 self-contained labs · exactly 4 labs per day
Assessment	WA-SAQ: 5 questions / 60 min + Case Study: 5 tasks / 60 min

Learning Outcomes

- LO1: Maintain business continuity of AWS cloud operations to ensure stable service delivery.
- LO2: Implement secure AWS cloud infrastructure setup such as data loading and performance monitoring.
- LO3: Apply service quality standards to ensure security and compliance of AWS cloud operations.
- LO4: Analyse and resolve processing issues to ensure continuity and disaster recovery in AWS cloud applications.
- LO5: Troubleshoot networking performance issues at scale for optimal AWS infrastructure operation.

Knowledge and Ability Statements

Code	Statement
K1	Cloud computing infrastructure components and technologies
K2	Installation, configuration and troubleshooting of cloud computing infrastructures
K3	Installation, configuration and troubleshooting of software packages supporting cloud computing infrastructures
K4	Design, configuration, maintenance and troubleshooting of cloud computing applications
K5	Virtualisation technologies
A1	Implement data loading, security, performance monitoring and tuning, capacity planning, backup and recovery
A2	Implement performance diagnostics and troubleshooting at scale in a live environment
A3	Maintain availability and performance of cloud computing operations and ensure secure services with no unplanned downtime

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A4	Analyse processing issues and data anomalies and identify solutions
A5	Implement and release cloud computing operations changes
A6	Implement IT risk-management measures for business continuity and disaster recovery
A7	Implement service-quality standards

Three-Day Learning Journey

Day 1 — Foundations, observability, EC2 and fleet operations

Start	End	Facilitated activity
09:00	09:45	Administration, outcomes and CloudOps transition
09:45	11:15	Topic 01 — Cloud Operations Foundations
11:15	13:00	Topic 02 — Monitoring and Incident Response + Labs 01–02
14:00	16:00	Topic 03 — EC2 and Systems Management + Lab 03
16:00	18:00	Topic 04 — Reliability and High Availability + Lab 04

Day 2 — Reliability, continuity, deployment automation and security

Start	End	Facilitated activity
09:00	11:00	Topic 05 — Backup and Continuity + Lab 06
11:00	13:00	Topic 06 — Provisioning and Automation + Lab 05
14:00	16:00	Topic 07 — Security and Compliance + Labs 07–08
16:00	18:00	Change, recovery and compliance synthesis across four labs

Day 3 — Networking, data operations, integrated troubleshooting and assessment

Start	End	Facilitated activity
09:00	11:00	Topic 08 — Networking, DNS and Content Delivery + Labs 09–10
11:00	13:00	Topic 09 — Storage and Database Operations + Lab 11
14:00	16:00	Topic 10 — Integrated Troubleshooting + Lab 12 capstone
16:00	18:00	Written Assessment (60 min) + Case Study (60 min)

Assessment Design

- Written Assessment (WA-SAQ) — five open-ended questions, 60 minutes, open book.
- Case Study (CS) — five connected tasks, 60 minutes, open book.
- Structure mirrors the Drive-held controlled papers and Assessment Plan v5.0: five WA questions covering K1–K5 and five Case Study tasks covering A1–A7.

Operational note: Answer keys are trainer-only. Candidate papers may be learner-facing through LMS-TMS; keys must not be published to LMS-TMS or public GitHub.

Source Register and Current Certification Context

Authority	Source	Use in this release
WSQ authority	Live Tertiary course page and	Course code, 24-hour duration, five

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	Assessment Plan v5.0	LO, K1–K5, A1–A7 and WA+CS design.
Legacy content	Reference v3/v4 PowerPoints and Drive Activities dataset	Preserved useful EC2, SSM, CloudFormation, storage, monitoring, security and networking coverage; procedures moved out of the deck.
Certification authority	AWS Certified CloudOps Engineer - Associate SOA-C03 Exam Guide v1.1	Current five-domain blueprint and scope, including containers, multi-account operations, automation and IaC.
Architecture guidance	AWS Well-Architected Framework and official AWS service documentation	Operational excellence, security, reliability, performance, cost and sustainability practices.
Assessment authority	Drive-held WA/CS v2 papers and Assessment Plan v5.0	Instrument type, five-item counts, 60-minute timings and controlled K/A coverage.

The certification role is now AWS Certified CloudOps Engineer — Associate (SOA-C03). The approved WSQ title and course code remain unchanged. This release uses the current five-domain exam guide for readiness while preserving approved WSQ learning outcomes and assessment design.

SOA-C03 domain	Weight
Monitoring, logging, analysis, remediation and performance optimisation	22%
Reliability and business continuity	22%
Deployment, provisioning and automation	22%
Security and compliance	16%
Networking and content delivery	18%

AWS Lab Environment and Safety

Control	Required practice
Identity	Confirm account alias or permitted account identifier, caller identity and Region before every deployment.
Ownership	Tag all resources with Course=TGS-2024051413 and a unique learner suffix.
Least privilege	Investigate denied actions; do not attach broad administrator policies as a shortcut.
Change boundary	Use canaries, change sets, bounded concurrency, health signals and rollback.
Cost	Use small lab resources, approved Regions and explicit cleanup.
Evidence	Save command output, timestamps, decisions, validation and cleanup results.
Secrets	Never commit credentials, tokens, private keys or full account identifiers.

```
aws sts get-caller-identity
aws configure get region
aws resourcegroupstaggingapi get-resources --tag-filters
Key=Course,Values=TGS-2024051413
```

Common Lab Troubleshooting

Symptom	Safe response
AccessDenied	Identify the API action and resource; check role, boundary, SCP and key policy before requesting scoped permission.
ValidationError	Check Region, parameters, resource state and unresolved placeholders.
No metric or log	Confirm namespace, dimensions, time range, integration and retention.
Health check fails	Follow DNS, route, security group, NACL, target and dependency in order.

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Cleanup fails	Preserve error output, identify retained dependencies and escalate immediately.

Cloud Operations Knowledge

Topic 01 — Cloud Operations Foundations

Alignment: LO1 · LO2 · K1 · K5 · A1 · Foundation across all five SOA-C03 domains. Shared responsibility · regions and zones · operational excellence · SOA-C03 transition

Key idea	Explanation
Operate for outcomes	Availability, security, performance and cost are managed as measurable service outcomes.
Know the responsibility boundary	AWS secures the cloud; customers secure configurations, identities, data and workloads in the cloud.
Design for failure domains	Regions, Availability Zones and edge locations create distinct resilience and latency choices.
Use evidence before change	Metrics, logs, configuration history and events establish the operational baseline.

SysOps to CloudOps — What Changed

Area	Legacy SOA-C02 emphasis	Current SOA-C03 emphasis
Role name	SysOps Administrator	CloudOps Engineer
Workloads	EC2-heavy operations	Compute, containers, serverless and hybrid operations
Scope	Six broad domains	Five task-oriented domains
Automation	AWS-native operations	IaC, Git and third-party automation are explicit

Operational note: The WSQ course title and code remain unchanged; exam-readiness content follows the current SOA-C03 guide.

The Cloud Operations Control Loop

Stage	Operational meaning
Observe	Signals and context
Assess	Impact and risk
Decide	Runbook and authority
Change	Small and reversible
Verify	Outcome and evidence
Learn	Prevent recurrence

Operational note: PROFESSIONAL STANDARD — Every operational change must have a reason, owner, verification signal and rollback path.

AWS Global Infrastructure

Concept	Explanation
Region	Geographic boundary for sovereignty, latency and disaster-recovery decisions.
Availability Zone	Independent failure domain connected with low-latency networking.
Edge	CloudFront and Route 53 move delivery and routing closer to users.
Local and hybrid	Outposts, Local Zones and connected environments extend the operating boundary.

Availability Improves with Independent Failure Domains

Category	Illustrative availability target (%)
Single instance	99.0
Two AZs	99.9
Three AZs	99.95

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Two Regions	99.99

Operational note: More failure domains increase potential resilience only when dependencies, health checks and failover are also designed correctly.

Shared Responsibility by Layer

Layer	AWS responsibility	Customer responsibility
Facilities and hardware	Physical security, hosts and core network	Choose Regions and resilience strategy
Managed service	Service infrastructure and platform	Identity, configuration, data and use
EC2 workload	Underlying facilities and hypervisor	Guest OS, patches, application and traffic controls
Data	Durable service mechanisms	Classification, encryption, retention and access

Well-Architected Operational Questions

Prepare: What must be observable?; What runbooks and permissions exist?.

Operate: What is normal?; Who responds to deviation?.

Evolve: What did evidence teach?; Which control prevents recurrence?.

Operations Mindset

The console is an interface—not the source of truth.

Operational note: Treat configuration, telemetry, change history and verification evidence as the durable operational record.

Topic 02 — Monitoring, Logging and Incident Response

Alignment: LO2 · LO4 · K1 · K3 · K4 · A1 · A4 · SOA-C03 Domain 1 · 22%. CloudWatch · CloudTrail · EventBridge · alarms · anomaly detection · remediation

Key idea	Explanation
Metrics show behaviour	Dimensions, periods and statistics turn raw measurements into service signals.
Logs explain events	Structured records provide detail for search, correlation, retention and audit.
Traces connect dependencies	A request path exposes latency and failure across distributed components.
Incidents require controlled response	Detection, triage, containment, recovery and learning form one evidence chain.

Telemetry Pipeline

Stage	Operational meaning
Instrument	Metrics, logs and traces
Collect	Agents and service integrations
Store	Retention and protection
Analyse	Queries and correlation
Alert	Actionable threshold
Remediate	Human or automation

Operational note: DESIGN TEST — A signal is useful only when it identifies impact, owner and next safe action.

CloudWatch, CloudTrail and Config

Service	Primary question	Typical evidence
CloudWatch	How is the workload behaving?	Metrics, logs, alarms and dashboards
CloudTrail	Who called which AWS API, when and from where?	Management and data events

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AWS Config	What configuration existed and was it compliant?	Configuration history and rule evaluation

Composite Signals Reduce Alert Noise

Category	Alerts per week	Actionable alerts
Raw threshold	84	9
Sustained breach	31	9
Composite alarm	14	8
Anomaly + impact	7	7

Operational note: Filtering transient and context-free signals can preserve actionable detection while reducing noise.

Alarm Design

Concept	Explanation
Statistic	Average can hide peaks; percentiles expose tail latency.
Period	Short periods detect quickly but may amplify noise.
Missing data	Choose breaching, not breaching, ignore or missing deliberately.
Action	Notify, scale or remediate only with tested permissions and rollback.

Incident Lifecycle

Stage	Operational meaning
Detect	Signal and timestamp
Triage	Scope and severity
Contain	Limit blast radius
Recover	Restore service
Verify	Health and data integrity
Review	Actions and owners

Operational note: A4 CONNECTION — Separate observed evidence, hypothesis, decision and result throughout the incident record.

Threshold vs Anomaly Detection

Choice	Strength	Use when
Static threshold	Transparent and predictable	A meaningful limit is known
Anomaly detection	Adapts to time-varying baseline	Seasonality makes one limit noisy
Composite alarm	Combines symptom and impact	One signal alone is insufficient
Metric math	Derives rates and ratios	Raw counts need context

Automation Guardrails

Trigger: Use sustained evidence; Prevent recursive events.

Authority: Least-privilege execution role; Separate approval for high-risk action.

Verification: Check intended state; Escalate when remediation fails.

Operational note: Applied in Lab 01 — Build an Observability and Incident Detection Baseline, Lab 02 — Automate Event-Driven Incident Triage.

Topic 03 — Compute, Virtualisation and Systems Management

Alignment: LO1 · LO2 · K3 · K5 · A3 · A5 · SOA-C03 Domains 1 and 3. EC2 lifecycle · AMIs · placement · Systems Manager · patching · Image Builder

Key idea	Explanation
Treat instances as managed nodes	Inventory, patch, access and automate consistently across a fleet.
Images are controlled supply-chain inputs	Build, scan, test and version AMIs instead of hand-

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	configuring servers.
Placement affects failure and performance	Spread, partition and cluster strategies solve different operational needs.
Lifecycle actions need coordination	Launch, stop, terminate and scale transitions require hooks and evidence.

EC2 Purchase and Capacity Models

Model	Operational value	Risk / constraint
On-Demand	Flexible baseline and unpredictable workloads	Higher unit cost
Savings Plans	Commitment discount across eligible usage	Forecast and commitment risk
Reserved capacity	Capacity assurance in an AZ	Scope and utilisation management
Spot	Large discount for interruptible work	Two-minute interruption and capacity variability

Golden Image Pipeline

Stage	Operational meaning
Source	Hardened base image
Build	Versioned components
Scan	Vulnerability and policy
Test	Boot and workload checks
Distribute	Accounts and Regions
Retire	Deprecate old images

Operational note: A5 CONTROL — An image release is an operational change with evidence, version and rollback.

Patch Compliance by Fleet Segment

Category	Compliant (%)	Managed-node coverage (%)
Web	96	100
Batch	92	98
Database tools	88	91
Legacy	67	72

Operational note: Patch compliance cannot exceed managed-node visibility; onboard and inventory the fleet before enforcing targets.

Systems Manager Capabilities

Concept	Explanation
Session Manager	Auditable shell access without inbound SSH or bastion exposure.
Run Command	Execute controlled commands across tagged targets.
Patch Manager	Baseline, maintenance window, compliance and remediation.
Automation	Runbooks coordinate multi-step operational changes.

Placement Strategies

Strategy	Optimises	Failure characteristic
Cluster	Low-latency high-throughput east-west traffic	Single-AZ concentration
Spread	Small number of critical instances	Distinct underlying hardware
Partition	Large distributed workloads	Partition-aware failure isolation

Safe Fleet Change

Stage	Operational meaning
Select	Tags and inventory
Canary	One representative node
Execute	Controlled concurrency
Observe	Health and errors
Expand	Waves by risk
Rollback	Known-good state

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Operational note: OPERATIONS RULE — Scale the change only after the canary evidence meets the acceptance threshold.

Fleet Principle

If a server cannot be identified, accessed, patched and replaced consistently, it is not operationally managed.

Operational note: Use tags, roles, inventory, image pipelines and Systems Manager to turn instances into a governed fleet.

Operational note: Applied in Lab 03 — Operate an EC2 Fleet with Systems Manager.

Topic 04 — Reliability, Elasticity and High Availability

Alignment: LO1 · K4 · K5 · A3 · A5 · SOA-C03 Domain 2 · 22%. Elastic Load Balancing · Auto Scaling · health checks · containers · graceful degradation

Key idea	Explanation
High availability is an end-to-end property	Compute redundancy fails if identity, data or network dependencies remain single points.
Elasticity follows demand	Auto Scaling changes capacity using a measurable signal and bounded policy.
Health checks drive routing	A check must represent the user's ability to receive a correct response.
Resilience requires recovery	Design degradation, retry, failover and rollback before an incident.

Load Balancer Selection

Type	Layer and strength	Typical workload
ALB	Layer 7 HTTP routing, host/path rules	Web, APIs and containers
NLB	Layer 4 performance and static IP	TCP/UDP/TLS and extreme scale
GWLB	Transparent appliance insertion	Firewalls and inspection fleets

Request Path and Health

Stage	Operational meaning
DNS	Resolve healthy endpoint
Load balancer	Select target
Target group	Health and port
Application	Dependency-aware response
Telemetry	Latency and errors
Scaling	Capacity response

Operational note: TROUBLESHOOTING — Follow the request path in order; do not jump directly to the instance.

Scaling Policy Behaviour

Category	Demand units	Capacity units
09:00	20	30
10:00	45	40
11:00	80	70
12:00	120	120
13:00	70	90
14:00	35	50

Operational note: Target tracking follows demand with delay; warm-up and cooldown prevent unstable oscillation.

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Auto Scaling Controls

Concept	Explanation
Desired capacity	The current fleet target between minimum and maximum bounds.
Target tracking	Adjusts capacity to hold a metric near a chosen target.
Step scaling	Uses breach magnitude to change capacity by defined steps.
Lifecycle hook	Pauses launch or termination for controlled work.

Failure Handling

Pattern	Response	Caution
Retry with backoff	Recover from transient dependency failure	Bound attempts and add jitter
Circuit breaker	Stop calling an unhealthy dependency	Define recovery probe
Bulkhead	Isolate resource pools	Capacity must reflect priority
Graceful degradation	Preserve critical function	Communicate reduced service

Container Operations in SOA-C03

ECS / EKS: Service desired count; Task or pod health and placement.

Images: Versioned and scanned; Immutable deployment input.

Observability: Container Insights; Logs, metrics and traces by workload.

Availability Test

Redundancy is not resilience until failover is exercised.

Operational note: Measure detection, decision, traffic shift, recovery, data integrity and return to steady state.

Operational note: Applied in Lab 04 — Deploy and Test a Multi-AZ Highly Available Service.

Topic 05 — Backup, Restore and Business Continuity

Alignment: LO1 · LO4 · K4 · A1 · A4 · A6 · SOA-C03 Domain 2 · 22%. RTO/RPO · AWS Backup · snapshots · replication · pilot light · warm standby

Key idea	Explanation
Backup is not recovery	A protected copy has value only when restore time and data integrity are proven.
RTO and RPO drive design	Recovery time and acceptable data loss determine technology and cost.
Cross-account protection limits blast radius	Separate vault ownership and controls protect against compromised workloads.
Continuity is a rehearsed capability	People, runbooks, dependencies and evidence matter as much as replication.

RTO and RPO

Objective	Question	Design influence
RTO	How quickly must service return?	Automation, standby capacity and failover
RPO	How much recent data may be lost?	Backup frequency and replication
Retention	How far back must recovery reach?	Lifecycle, vault and compliance
Integrity	How do we know restored data is correct?	Checksums and application validation

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Continuity Patterns Trade Cost for Recovery

Category	Relative steady-state cost	Recovery speed index
Backup/restore	1	2
Pilot light	3	5
Warm standby	6	8
Active/active	10	10

Operational note: Choose the least costly pattern that demonstrably meets business RTO, RPO and compliance requirements.

Recovery Assurance Loop

Stage	Operational meaning
Classify	Workload and data
Protect	Plan and vault
Monitor	Job and copy status
Restore	Isolated target
Validate	Application and data
Improve	Gap and owner

Operational note: A6 EVIDENCE — A completed backup job is an input; a tested restore with measured recovery is evidence.

AWS Backup Controls

Concept	Explanation
Plan	Schedules, windows, lifecycle and copy actions.
Vault	Access policy, encryption and logical separation.
Vault Lock	Retention controls designed to resist early deletion.
Audit Manager	Evidence against backup policy and compliance frameworks.

Replication Choices

Mechanism	Strength	Operational concern
EBS snapshot copy	Point-in-time regional protection	Restore and launch workflow
S3 replication	Object-level asynchronous copy	Existing objects and delete markers
RDS cross-Region replica	Lower RPO read copy	Promotion and endpoint change
DynamoDB global table	Multi-Region active data	Conflict and application behaviour

Continuity Runbook

Decision: Declared trigger and authority; Communication path.

Execution: Dependency order; Automated and manual steps.

Evidence: Timestamps and validation; Actual RTO/RPO and gaps.

Recovery Principle

A restore you have not tested is only a hypothesis.

Operational note: Schedule recovery exercises, isolate tests from production, measure actual results and close every gap with an owner.

Operational note: Applied in Lab 06 — Prove Backup and Restore Readiness.

Topic 06 — Deployment, Provisioning and Automation

Alignment: LO2 · LO4 · K3 · K4 · A4 · A5 · SOA-C03 Domain 3 · 22%. CloudFormation · StackSets · change sets · drift · Systems Manager Automation · Git and IaC

Key idea	Explanation

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Infrastructure is versioned	Templates and policies create a reviewable desired state.
Preview before execution	Change sets reveal creates, replacements and deletions before the stack changes.
Drift is a control signal	Difference between desired and actual state requires investigation, not blind overwrite.
Automation must be idempotent	Repeated execution should converge safely to the intended state.

Infrastructure Change Path

Stage	Operational meaning
Branch	Versioned proposal
Validate	Syntax and policy
Plan	Change set
Approve	Risk and authority
Apply	Bounded deployment
Verify	State, service and rollback

Operational note: A5 CONNECTION — The template, plan, approval, execution and verification form one auditable release record.

CloudFormation Update Controls

Control	Purpose	Operational use
Change set	Preview resource actions	Review replacements and deletions
Stack policy	Protect critical resources	Restrict updates during normal change
Rollback	Return after failed update	Preserve events for diagnosis
Termination protection	Prevent accidental stack deletion	Use on persistent environments

Automation Reduces Configuration Variance

Category	Configuration variance index	Change evidence score
Manual	10	2
Scripted	6	5
IaC	3	8
IaC + policy	1	10

Operational note: Automation improves consistency; policy and version control make the change explainable and governable.

Template Engineering

Concept	Explanation
Parameters	Expose safe environment-specific values without duplicating templates.
Mappings and conditions	Represent controlled variation, not arbitrary complexity.
Outputs and exports	Publish stable integration values with dependency awareness.
Deletion and update policies	Protect or snapshot stateful resources during change.

StackSets Deployment Models

Model	Best fit	Trade-off
Self-managed	Explicit administration and target roles	More role setup and control
Service-managed	AWS Organizations integration	Requires trusted access and OU planning
Automatic deployment	New accounts inherit baseline	Test policy and Region scope carefully

Automation Runbook Design

Inputs: Validated parameters; No embedded secrets.

Control: Retries and timeouts; Approval for destructive steps.

Output: Structured evidence; Explicit success and rollback.

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IaC Discipline

Do not fix drift by hiding it.

Operational note: Identify who changed what and why, decide whether actual or desired state is correct, then reconcile through the controlled source.

Operational note: Applied in Lab 05 — Control Infrastructure Change and Configuration Drift.

Topic 07 — Security, Identity and Compliance

Alignment: LO2 · LO3 · K1 · K3 · A1 · A7 · SOA-C03 Domain 4 · 16%. IAM · Organizations · KMS · Config · Security Hub · GuardDuty · Inspector · Macie

Key idea	Explanation
Identity is the control plane	Short-lived credentials, least privilege and explicit trust bound administrative power.
Prevent, detect and respond	Policies prevent known bad states; telemetry detects deviation; runbooks contain and recover.
Encryption includes key operations	Access, rotation, grants, aliases and recovery must be governed—not only enabled.
Compliance requires evidence	A rule result is useful when linked to scope, owner, remediation and exception process.

Identity Policy Layers

Layer	What it controls	Common failure
Identity policy	Permissions granted to a principal	Overbroad action or resource
Resource policy	Who may access a resource	Unintended external principal
Permission boundary	Maximum identity permissions	Assuming it grants access
SCP	Maximum permissions in an organisation	Blocking required service operation

Access Evaluation

Stage	Operational meaning
Authenticate	Principal and session
Context	Action, resource and conditions
Explicit deny	Any matching deny wins
Allow	At least one applicable allow
Boundary	Maximum permitted set
Decision	Allow or deny with evidence

Operational note: TROUBLESHOOTING — Use CloudTrail, policy simulator and access analysis to explain the decision path.

Security Findings Need Prioritisation

Category	Open findings	SLA hours
Critical	3	4
High	14	24
Medium	37	120
Low	82	720

Operational note: Severity, exposure, asset value and exploitability drive response; high volume alone does not determine priority.

Security Services by Question

Concept	Explanation
GuardDuty	What suspicious account, workload or network activity is occurring?
Inspector	Which compute workloads have vulnerabilities or exposure?

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Macie	Where is sensitive data in S3 and how is it exposed?
Security Hub	How do findings and standards combine across accounts?

KMS Key Choices

Choice	Operational control	Use carefully
AWS owned	Service manages lifecycle	Limited customer control
AWS managed	Visible service key	Policy is service-oriented
Customer managed	Policy, grants, rotation and aliases	Availability and deletion governance
Multi-Region key	Related key material across Regions	Application and recovery design

Compliance Remediation

Stage	Operational meaning
Define	Required state
Evaluate	Config rule or control
Prioritise	Risk and scope
Remediate	Automation or owner
Verify	Compliant state
Exception	Approved expiry and review

Operational note: A7 OUTPUT — Service quality is a measurable standard with evidence, not a one-time checkbox.

Security Operations

Least privilege is a lifecycle, not a launch-day setting.

Operational note: Observe actual use, remove unused permissions, constrain trust, investigate denied actions and revalidate after organisational change.

Operational note: Applied in Lab 07 — Build a Multi-Account Compliance Evidence Flow, Lab 08 — Diagnose IAM Access and Protect Data with KMS.

Topic 08 — Networking, DNS and Content Delivery

Alignment: LO5 · K2 · A2 · SOA-C03 Domain 5 · 18%. VPC · routes · security groups · NACLs · endpoints · Route 53 · CloudFront

Key idea	Explanation
Connectivity is a path	DNS, route, policy, state, endpoint and application health must all succeed.
Security groups are stateful	Return traffic is automatically allowed; network ACLs evaluate each direction.
Private connectivity reduces exposure	VPC endpoints, Transit Gateway and hybrid links change routing and policy boundaries.
DNS is a control surface	Routing policy, health evaluation, TTL and resolver behaviour affect recovery.

Connectivity Troubleshooting Order

Stage	Operational meaning
Name	DNS answer and TTL
Route	Longest prefix and target
Policy	SG, NACL and endpoint policy
State	NAT, connection and asymmetric path
Target	Listener, port and health
Evidence	Flow logs and application logs

Operational note: A2 STANDARD — Test from source to destination and record the first failed control boundary.

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Security Group vs Network ACL

Property	Security group	Network ACL
Scope	Elastic network interface	Subnet
State	Stateful	Stateless
Rules	Allow only	Numbered allow and deny
Evaluation	All rules together	Lowest rule number first

DNS TTL Trades Agility for Query Load

Category	Failover agility index	Resolver query load index
30 sec	10	10
60 sec	8	7
300 sec	4	3
900 sec	2	1

Operational note: Lower TTL can speed client refresh but increases query volume and cannot force every resolver to behave identically.

Route 53 Routing Policies

Concept	Explanation
Simple	One or more values without health-based policy logic.
Weighted	Controlled traffic split for migration or experiments.
Latency	Direct users to the Region with lowest measured latency.
Failover	Primary and secondary routing driven by health.

VPC Endpoint Types

Endpoint	Connectivity	Common use
Gateway	Route-table target	S3 and DynamoDB
Interface	Private IP through PrivateLink	Many AWS and partner services
Gateway Load Balancer	Inspection appliance insertion	Central security services
Resource / service network	VPC Lattice private application access	Cross-VPC service connectivity

CloudFront Operations

Cache: Cache key and TTL; Invalidation and versioned objects.

Origin: Origin access control; Origin health and failover.

Security: TLS certificate and WAF; Logs and real-time metrics.

Network Diagnosis

A timeout is a symptom, not a location.

Operational note: Use DNS results, route tables, reachability analysis, flow logs, load-balancer health and application evidence to isolate the boundary.

Operational note: Applied in Lab 09 — Troubleshoot a VPC Connectivity Path, Lab 10 — Operate DNS and Content-Delivery Failover.

Topic 09 — Storage, Data and Database Operations

Alignment: LO1 · LO2 · LO4 · K1 · K4 · A1 · A4 · A6 · Cross-domain operations. S3 · EBS · EFS · RDS · DynamoDB · lifecycle · performance · data protection

Key idea	Explanation
Choose by access pattern	Object, block, file and database services solve different consistency, latency and sharing needs.
Performance has service-specific limits	IOPS, throughput, connections, partitions and caching require different evidence.

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Data protection is layered	Encryption, versioning, snapshots, backup and replication address different risks.
Managed does not mean unmanaged	Customers still own schema, configuration, access, capacity and recovery choices.

Storage Selection

Service	Access model	Operational fit
S3	Object and API	Durable data lake, backup and static content
EBS	Block device to compute	Low-latency instance and database volumes
EFS	Shared NFS file system	Elastic multi-instance Linux file access
FSx	Managed specialised file system	Windows, Lustre, NetApp or OpenZFS workloads

S3 Data Protection

Stage	Operational meaning
Block	Public access controls
Authorise	Identity and bucket policy
Encrypt	Default SSE and KMS
Version	Recover overwrite or delete
Lifecycle	Transition and expire
Replicate	Account or Region copy

Operational note: A1 CONNECTION — Protection combines preventive access, recoverability, retention and evidence.

EBS Performance Must Match Workload

Category	Relative IOPS	Relative throughput
gp3 baseline	3	3
gp3 tuned	8	6
io2	10	5
st1	1	10

Operational note: IOPS and throughput are different dimensions; select volume type and configuration from measured workload behaviour.

RDS Availability and Scale

Feature	Primary goal	Operational note
Multi-AZ	Availability and failover	Standby is not a read-scaling target
Read replica	Read scale and reporting	Replication lag and promotion plan
Automated backup	Point-in-time recovery	Retention and restore testing
RDS Proxy	Connection pooling and resilience	Application compatibility and limits

DynamoDB Operations

Concept	Explanation
Capacity	On-demand or provisioned with auto scaling and adaptive capacity.
Partition key	Distribute traffic; avoid hot keys and unbounded item growth.
Recovery	Point-in-time recovery, on-demand backup and restore.
Global table	Multi-Region active access with application-aware conflict behaviour.

Database Incident Evidence

Resource: CPU, memory and storage; Connections and queue depth.

Engine: Slow queries and locks; Wait events and plans.

Change: Configuration and schema history; Deployment and failover events.

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Data Operations

Durability claims do not replace recovery design.

Operational note: Protect the correct scope, control access, monitor jobs, test restore, validate application behaviour and measure actual RTO/RPO.

Operational note: Applied in Lab 11 — Diagnose Storage and Database Performance, Then Recover.

Topic 10 — Performance, Cost and Integrated Troubleshooting

Alignment: LO3 · LO4 · LO5 · K2 · K4 · A2 · A4 · A7 · Integration across SOA-C03 domains. Bottleneck analysis · Cost Explorer · Trusted Advisor · runbooks · game days · capstone

Key idea	Explanation
Optimise after measurement	Right-size and tune using utilisation, saturation, latency and business demand.
Cost anomalies are operational signals	Unexpected spend can reveal leakage, misconfiguration, attack or failed cleanup.
Troubleshoot hypotheses	A runbook narrows evidence systematically instead of changing multiple variables.
Service quality is verified	SLOs, recovery tests and post-incident actions prove operational capability.

Evidence-Led Troubleshooting

Stage	Operational meaning
Frame	Impact, scope and time
Baseline	Expected behaviour
Correlate	Signals and recent change
Hypothesise	Ranked possible causes
Test	One safe discriminating action
Verify	Outcome and recurrence control

Operational note: A4 STANDARD — A hypothesis is not a cause until evidence supports it and alternatives are tested.

Performance Signals

Signal	Interpretation	Example action
Utilisation	Resource use relative to capacity	Right-size only with sustained evidence
Saturation	Queued work or constrained resource	Increase capacity or remove bottleneck
Errors	Failed operations or retries	Trace dependency and change history
Latency	Time experienced by request	Inspect percentiles and request path

Cost Anomaly Investigation

Category	Baseline daily cost	Observed daily cost
Compute	220	235
Storage	90	92
Data transfer	45	180
Database	130	135
Logging	25	70

Operational note: The largest percentage changes are data transfer and logging; investigate traffic path, retention and recent configuration before right-sizing compute.

Cost and Optimisation Tools

Concept	Explanation
Cost Explorer	Trend, filter, group and forecast historical spend.

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Cost Anomaly Detection	Surface spend patterns that deviate from expected behaviour.
Compute Optimizer	Resource recommendations from utilisation history.
Trusted Advisor	Checks across cost, security, resilience, performance and limits.

Runbook vs Playbook

Artifact	Purpose	Contents
Runbook	Repeatable technical procedure	Inputs, steps, verification and rollback
Playbook	Coordinated response to a scenario	Roles, decisions, communication and runbooks
Game day	Exercise assumptions safely	Scenario, injects, observations and improvements
Post-incident review	Prevent recurrence	Timeline, causes, actions and verification

Service Quality Review

Reliability: SLO and error budget; Recovery test results.

Security: Findings and exceptions; Control effectiveness.

Efficiency: Capacity and cost; Automation and toil reduction.

Capstone Standard

Restore service, explain the evidence, and improve the system.

Operational note: A complete CloudOps response connects detection, diagnosis, controlled recovery, validation, communication and recurrence prevention.

Operational note: Applied in Lab 12 — Capstone — Resolve a Multi-Service CloudOps Incident.

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Detailed AWS Lab Guides

Lab 01 — Build an Observability and Incident Detection Baseline

Field	Lab specification
Day / topic	Day 1 · Topic 02
Objective	Apply K1, K3, K4, A1 and A4 by creating a dashboard, alarm, CloudTrail evidence query and EventBridge detection rule for one tagged workload.
Scenario	Northstar Retail has intermittent web latency but no agreed baseline, owner or repeatable incident-evidence process.
Deliverable	A CloudWatch dashboard, alarm definition, EventBridge pattern, evidence log and validation screenshots.
Services	CloudWatch · CloudTrail · EventBridge · SNS · AWS CLI · CloudShell
Visual	EXPECTED-ARCHITECTURE.jpg

Before you begin

- Confirm caller identity and Region.
- Read every supplied editable YAML, JSON, shell, Python or CLI asset.
- Resolve every REPLACE placeholder and use a unique learner suffix.
- Define expected service, security, performance and cleanup evidence before changing AWS.

Step-by-step procedure

Step 1 — Open `LAB-PLAN.md`; record the assigned Region, workload tag, notification email and cost boundary before creating resources.

```
open LAB-PLAN.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 2 — Confirm the active identity and Region in CloudShell. Save the output in `evidence/caller-identity.txt`.

```
aws sts get-caller-identity | tee evidence/caller-identity.txt
aws configure get region | tee evidence/region.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 3 — Create or select a small tagged EC2 training workload and confirm the `Course=TGS-2024051413` tag is visible.

```
aws ec2 describe-instances --filters 'Name=tag:Course,Values=TGS-2024051413' --query
'Reservations[].Instances[].
{Id:InstanceId,State:State.Name,AZ:Placement.AvailabilityZone}' --output table
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

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Step 4 — Replace the placeholder instance ID in `cloudwatch-dashboard.json`, then create the dashboard.

```
aws cloudwatch put-dashboard --dashboard-name SysOps-Lab01 --dashboard-body
file://cloudwatch-dashboard.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 5 — Replace the placeholder values in `alarm.json`; create the CPU alarm and verify its current state.

```
aws cloudwatch put-metric-alarm --cli-input-json file://alarm.json
aws cloudwatch describe-alarms --alarm-names SysOps-Lab01-HighCPU
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 6 — Review CloudTrail Event history for your identity and export recent management events.

```
aws cloudtrail lookup-events --max-results 20 --query 'Events[].
{Time:EventTime,Name:EventName,User:Username}' --output table | tee
evidence/cloudtrail-events.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 7 — Create the EventBridge rule from `event-pattern.json`; add the trainer-approved SNS target only after subscription confirmation.

```
aws events put-rule --name SysOps-Lab01-ConsoleChange --event-pattern file://event-
pattern.json --state ENABLED
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 8 — Generate one safe configuration event, locate it in CloudTrail and confirm the EventBridge rule metric changes.

```
aws events describe-rule --name SysOps-Lab01-ConsoleChange | tee evidence/event-
rule.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 9 — Complete `INCIDENT-EVIDENCE.md`: separate observation, hypothesis, impact, next test and owner.

```
open INCIDENT-EVIDENCE.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

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Step 10 — Capture the dashboard, alarm and event rule as JPEG screenshots; compare them with `EXPECTED-ARCHITECTURE.jpg`.

```
open EXPECTED-ARCHITECTURE.jpg
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 11 — Run `validate.sh`; resolve missing placeholders or evidence before cleanup.

```
bash validate.sh
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 12 — Delete the lab rule, alarm and dashboard unless the trainer approves reuse; record cleanup evidence.

```
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Verification

The dashboard and alarm reference the assigned workload; CloudTrail evidence names the actor and event; the EventBridge pattern is valid; the incident record distinguishes observation from hypothesis; validation emits LAB 01 PASS.

Evidence to retain

- Dashboard JPEG
- Alarm JPEG
- CloudTrail event export
- EventBridge rule export
- completed INCIDENT-EVIDENCE.md
- cleanup.txt

Acceptance criteria

- No placeholder values remain
- All resources carry the course tag where supported
- Alarm has an owner and action
- Evidence includes UTC timestamps
- No unapproved paid resource remains

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Lab 02 — Automate Event-Driven Incident Triage

Field	Lab specification
Day / topic	Day 1 · Topic 02
Objective	Apply K3, K4, A1 and A4 by routing a controlled AWS event through EventBridge, invoking a least-privilege remediation function and preserving an auditable triage record.
Scenario	Northstar Retail needs to detect an unapproved security-group change quickly, enrich the event and notify the operator without making an unsafe automatic network change.
Deliverable	An EventBridge rule, Lambda triage function, test event, CloudWatch Logs evidence and incident decision record.
Services	EventBridge · Lambda · CloudTrail · CloudWatch Logs · IAM
Visual	EXPECTED-ARCHITECTURE.jpg

Before you begin

- Confirm caller identity and Region.
- Read every supplied editable YAML, JSON, shell, Python or CLI asset.
- Resolve every REPLACE placeholder and use a unique learner suffix.
- Define expected service, security, performance and cleanup evidence before changing AWS.

Step-by-step procedure

Step 1 — Review the architecture JPEG and `event-pattern.json`; identify event source, API names, principal context and target.

```
python3 -m json.tool event-pattern.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 2 — Inspect `triage.py`; confirm it records event identity and never modifies the security group.

```
python3 -m py_compile triage.py
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 3 — Create the trainer-approved execution role from `lambda-role-policy.json`; replace only the account-specific trust and resource values.

```
python3 -m json.tool lambda-role-policy.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 4 — Package and create the Lambda function with a unique learner suffix.

```
zip -j triage.zip triage.py
aws lambda create-function --function-name SysOps-Lab02-Triage-REPLACE_SUFFIX --
runtime python3.13 --handler triage.handler --zip-file fileb://triage.zip --role
REPLACE_ROLE_ARN
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 5 — Create the EventBridge rule and add the Lambda target; grant EventBridge permission to invoke only this function.

```
aws events put-rule --name SysOps-Lab02-SGChange --event-pattern file://event-pattern.json --state ENABLED
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 6 — Invoke the function with `test-event.json`; save the response and locate the matching CloudWatch Logs event.

```
aws lambda invoke --function-name SysOps-Lab02-Triage-REPLACE_SUFFIX --payload fileb://test-event.json evidence/invoke-response.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 7 — Generate one trainer-approved learner-owned security-group change, then verify the real event arrives through EventBridge.

```
aws logs tail /aws/lambda/SysOps-Lab02-Triage-REPLACE_SUFFIX --since 10m | tee evidence/triage-log.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 8 — Complete `TRIAGE-DECISION.md`: impact, evidence, safe next action, authority, verification and escalation.

```
open TRIAGE-DECISION.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 9 — Capture EventBridge rule, Lambda monitor and log evidence as JPEG screenshots.

```
open EXPECTED-ARCHITECTURE.jpg
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 10 — Run validation; delete the rule, target, permission, function and learner-owned test change.

```
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Verification

The test and real events reach the Lambda target, logs contain the event ID and actor context, no automatic network modification occurs, the decision record names authority and verification, and validation emits LAB 02 PASS.

Evidence to retain

- EventBridge rule export
- Lambda invocation response
- CloudWatch Logs export
- JPEG screenshots
- completed TRIAGE-DECISION.md
- cleanup.txt

Acceptance criteria

- Event pattern is limited to named API calls
- Lambda role is least privilege
- Function performs triage only
- Event ID and UTC time are preserved
- Cleanup removes all learner-created automation

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Lab 03 — Operate an EC2 Fleet with Systems Manager

Field	Lab specification
Day / topic	Day 1 · Topic 03
Objective	Apply K3, K5, A3 and A5 by onboarding a managed node, using Session Manager and Run Command, assessing patch compliance and executing a controlled Automation runbook.
Scenario	Northstar Retail must remove public SSH access and make patching repeatable across tagged web instances.
Deliverable	An EC2 managed node, SSM document, patch-compliance report, automation execution record and rollback evidence.
Services	EC2 · IAM · Systems Manager · Session Manager · Run Command · Patch Manager · Automation
Visual	EXPECTED-ARCHITECTURE.jpg

Before you begin

- Confirm caller identity and Region.
- Read every supplied editable YAML, JSON, shell, Python or CLI asset.
- Resolve every REPLACE placeholder and use a unique learner suffix.
- Define expected service, security, performance and cleanup evidence before changing AWS.

Step-by-step procedure

Step 1 — Review `managed-instance.yaml`; choose the trainer-provided VPC/subnet and confirm the AMI is supported in the assigned Region.

```
aws cloudformation validate-template --template-body file://managed-instance.yaml
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 2 — Deploy the stack using a unique learner suffix and the required IAM capability.

```
aws cloudformation deploy --stack-name SysOps-Lab03-REPLACE_SUFFIX --template-file
managed-instance.yaml --capabilities CAPABILITY_NAMED_IAM --parameter-overrides
CourseTag=TGS-2024051413
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 3 — Obtain the instance ID from stack outputs and wait until Systems Manager reports it online.

```
aws ssm describe-instance-information --filters Key=tag:Course,Values=TGS-2024051413
--output table | tee evidence/managed-nodes.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 4 — Start a Session Manager session. Confirm there is no inbound SSH rule and capture the managed shell evidence.

```
aws ssm start-session --target REPLACE_INSTANCE_ID
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 5 — Create the custom SSM document from `document-install-healthcheck.yml` and inspect its status.

```
aws ssm create-document --name SysOps-Lab03-HealthCheck --document-type Command --
document-format YAML --content file://document-install-healthcheck.yml
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 6 — Run the document against the managed node with a concurrency of one and record the command ID.

```
aws ssm send-command --document-name SysOps-Lab03-HealthCheck --targets
Key=tag:Course,Values=TGS-2024051413 --max-concurrency 1 --max-errors 0 | tee
evidence/send-command.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 7 — Review Inventory and patch compliance. Explain the baseline, severity and approval rules in `CHANGE-RECORD.md`.

```
aws ssm list-compliance-items --resource-ids REPLACE_INSTANCE_ID --resource-types
ManagedInstance --output table | tee evidence/compliance.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 8 — Execute the read-only Automation runbook and save the output. Do not scale beyond one node until its evidence passes.

```
aws ssm start-automation-execution --document-name AWS-DescribeInstanceInformation
--parameters InstanceIds=REPLACE_INSTANCE_ID | tee evidence/automation.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 9 — Capture Fleet Manager, compliance and command output as JPEG evidence; compare with `EXPECTED-ARCHITECTURE.jpg`.

```
open EXPECTED-ARCHITECTURE.jpg
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 10 — Complete the change record with risk, owner, success metric, rollback and UTC timestamps.

```
open CHANGE-RECORD.md
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 11 — Run `validate.sh`, then remove the custom document and CloudFormation stack.

```
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Verification

The instance is managed without public SSH, the custom document validates, command and compliance evidence are present, the change record includes verification and rollback, and validation emits LAB 03 PASS.

Evidence to retain

- Managed-node inventory
- Session Manager JPEG
- Run Command output
- Patch compliance export
- Automation execution
- completed CHANGE-RECORD.md
- cleanup.txt

Acceptance criteria

- No inbound SSH from the internet
- Instance profile grants only required managed-node permissions
- Canary concurrency is one
- Patch status is explained
- Rollback and cleanup are documented

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Lab 04 — Deploy and Test a Multi-AZ Highly Available Service

Field	Lab specification
Day / topic	Day 1 · Topic 04
Objective	Apply K4, K5, A3 and A5 by deploying an ALB and Auto Scaling service across two Availability Zones, testing health-based recovery and recording a controlled change.
Scenario	Northstar Retail's single-instance web service cannot survive an instance or Availability Zone disruption.
Deliverable	A two-AZ VPC web tier, ALB, target group, launch template, Auto Scaling group, failover evidence and cleanup record.
Services	VPC · EC2 · ALB · Auto Scaling · CloudWatch · CloudFormation
Visual	EXPECTED-ARCHITECTURE.jpg

Before you begin

- Confirm caller identity and Region.
- Read every supplied editable YAML, JSON, shell, Python or CLI asset.
- Resolve every REPLACE placeholder and use a unique learner suffix.
- Define expected service, security, performance and cleanup evidence before changing AWS.

Step-by-step procedure

Step 1 — Review `ha-web-stack.yaml` and the architecture JPEG. Identify the two failure domains, health-check path, minimum capacity and rollback boundary.

```
open EXPECTED-ARCHITECTURE.jpg
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 2 — Validate the template and deploy it with a unique suffix.

```
aws cloudformation validate-template --template-body file://ha-web-stack.yaml
aws cloudformation deploy --stack-name SysOps-Lab04-REPLACE_SUFFIX --template-file
ha-web-stack.yaml --capabilities CAPABILITY_NAMED_IAM
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 3 — Read the stack outputs; save the ALB URL, target group and Auto Scaling group identifiers.

```
aws cloudformation describe-stacks --stack-name SysOps-Lab04-REPLACE_SUFFIX --query
'Stacks[0].Outputs' --output table | tee evidence/stack-outputs.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 4 — Open the ALB URL repeatedly and confirm both Availability Zones contain healthy targets.

```
aws elbv2 describe-target-health --target-group-arn REPLACE_TARGET_GROUP_ARN --
output table | tee evidence/target-health-before.txt
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 5 — Inspect desired, minimum and maximum capacity. Confirm the group is configured across two subnets.

```
aws autoscaling describe-auto-scaling-groups --auto-scaling-group-names
REPLACE_ASG_NAME --output json | tee evidence/asg-before.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 6 — Run the bounded failure test by terminating one learner-owned instance in the Auto Scaling group; record the UTC start time and change ID.

```
aws ec2 terminate-instances --instance-ids REPLACE_CANARY_INSTANCE_ID | tee
evidence/termination.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 7 — Observe target health, replacement activity and ALB availability until desired capacity and health recover.

```
bash failover-observe.sh REPLACE_ASG_NAME REPLACE_TARGET_GROUP_ARN | tee
evidence/failover-observation.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 8 — Calculate detection and recovery time in `FAILOVER-REPORT.md`; explain whether the observed result meets the stated objective.

```
open FAILOVER-REPORT.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 9 — Create the target-tracking policy from `scaling-policy.json` only if the trainer approves a scaling demonstration.

```
aws autoscaling put-scaling-policy --cli-input-json file://scaling-policy.json | tee
evidence/scaling-policy.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 10 — Capture ALB health, Auto Scaling activity and CloudWatch signals as JPEG evidence.

```
open EXPECTED-ARCHITECTURE.jpg
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 11 — Run validation, delete the stack and wait for DELETE_COMPLETE before leaving the lab.

```
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Verification

The service spans two Availability Zones, returns a healthy response during one-instance replacement, recovers desired capacity, records measured recovery time, and validation emits LAB 04 PASS.

Evidence to retain

- Stack outputs
- Target health before/after
- Auto Scaling activity
- ALB response JPEG
- completed FAILOVER-REPORT.md
- cleanup.txt

Acceptance criteria

- Two distinct Availability Zones
- Minimum and desired capacity at least two
- Health check represents application response
- Failure test affects only learner-owned canary
- Measured recovery and rollback are documented

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Lab 05 — Control Infrastructure Change and Configuration Drift

Field	Lab specification
Day / topic	Day 2 · Topic 06
Objective	Apply K3, K4, A4 and A5 by validating a CloudFormation template, reviewing a change set, detecting drift and using a Systems Manager Automation runbook with verification.
Scenario	A manual console change has made Northstar Retail's production-like environment differ from its approved template.
Deliverable	A validated CloudFormation stack, reviewed change set, drift report, reconciliation decision, Automation verification and cleanup record.
Services	CloudFormation · Change Sets · Drift Detection · Systems Manager Automation · AWS CLI
Visual	EXPECTED-ARCHITECTURE.jpg

Before you begin

- Confirm caller identity and Region.
- Read every supplied editable YAML, JSON, shell, Python or CLI asset.
- Resolve every REPLACE placeholder and use a unique learner suffix.
- Define expected service, security, performance and cleanup evidence before changing AWS.

Step-by-step procedure

Step 1 — Validate `operations-stack.yaml` locally and with CloudFormation; inspect parameters, tags, deletion policy and output.

```
aws cloudformation validate-template --template-body file://operations-stack.yaml |
tee evidence/template-validation.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 2 — Deploy the baseline stack and record its stack ID and output.

```
aws cloudformation deploy --stack-name SysOps-Lab05-REPLACE_SUFFIX --template-file
operations-stack.yaml --parameter-overrides Environment=lab
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 3 — Edit only the working copy as instructed, then create—not execute—the change set.

```
aws cloudformation create-change-set --stack-name SysOps-Lab05-REPLACE_SUFFIX --
change-set-name Lab05-ReviewedChange --template-body file://operations-stack.yaml --
change-set-type UPDATE --parameters ParameterKey=Environment,ParameterValue=lab |
tee evidence/change-set-create.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 4 — Describe the change set. Record every Add, Modify, Remove or Replacement action in `CHANGE-DECISION.md`.

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```
aws cloudformation describe-change-set --stack-name SysOps-Lab05-REPLACE_SUFFIX --
change-set-name Lab05-ReviewedChange | tee evidence/change-set-review.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 5 — Execute the trainer-approved change set and wait for UPDATE_COMPLETE.

```
aws cloudformation execute-change-set --stack-name SysOps-Lab05-REPLACE_SUFFIX --
change-set-name Lab05-ReviewedChange
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 6 — Create one safe manual tag drift on the learner-owned resource; record actor and UTC time.

```
aws s3api put-bucket-tagging --bucket REPLACE_BUCKET --tagging
'TagSet=[{Key=Course,Value=TGS-2024051413},{Key=DriftDemo,Value=manual}]'
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 7 — Start stack drift detection, wait for completion and export the resource drift detail.

```
aws cloudformation detect-stack-drift --stack-name SysOps-Lab05-REPLACE_SUFFIX | tee
evidence/drift-detection.json
aws cloudformation describe-stack-resource-drifts --stack-name SysOps-Lab05-
REPLACE_SUFFIX | tee evidence/drift-details.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 8 — Decide whether desired or actual state is correct. Reconcile through the template rather than hiding the drift.

```
open CHANGE-DECISION.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 9 — Create and execute `automation-runbook.yaml` against the learner-owned bucket; save the execution output.

```
aws ssm create-document --name SysOps-Lab05-VerifyBucket --document-type Automation
--document-format YAML --content file://automation-runbook.yaml
aws ssm start-automation-execution --document-name SysOps-Lab05-VerifyBucket --
parameters BucketName=REPLACE_BUCKET | tee evidence/automation.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 10 — Capture change-set and drift evidence as JPEG screenshots, run validation, then clean up.

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```
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Verification

The change set is reviewed before execution, drift evidence identifies the changed property, the decision record explains reconciliation, the Automation runbook verifies state, and validation emits LAB 05 PASS.

Evidence to retain

- Template validation
- Change-set review
- Drift details
- Automation execution
- completed CHANGE-DECISION.md
- JPEG evidence
- cleanup.txt

Acceptance criteria

- No direct unreviewed production-style change
- Replacement actions explicitly reviewed
- Manual drift is confined to learner-owned resource
- Reconciliation decision cites evidence
- Automation has bounded permissions and explicit output

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Lab 06 — Prove Backup and Restore Readiness

Field	Lab specification
Day / topic	Day 2 · Topic 05
Objective	Apply K4, A1, A4 and A6 by protecting an S3/EBS-style workload, restoring into isolation and measuring actual recovery evidence.
Scenario	Northstar Retail reports successful backup jobs but has never proven that the protected data can be restored within its business objectives.
Deliverable	A backup plan, selection, restore checklist, recovery evidence and measured RTO/RPO report.
Services	AWS Backup · S3 · EBS · KMS · CloudWatch Events
Visual	EXPECTED-ARCHITECTURE.jpg

Before you begin

- Confirm caller identity and Region.
- Read every supplied editable YAML, JSON, shell, Python or CLI asset.
- Resolve every REPLACE placeholder and use a unique learner suffix.
- Define expected service, security, performance and cleanup evidence before changing AWS.

Step-by-step procedure

Step 1 — Read `RECOVERY-OBJECTIVES.md`; agree the workload scope, RTO, RPO, retention and restore-validation method.

```
open RECOVERY-OBJECTIVES.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 2 — Validate `backup-plan.json` and replace the learner-specific vault and role values.

```
python3 -m json.tool backup-plan.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 3 — Create the backup vault and plan with the trainer-approved encryption key and retention.

```
aws backup create-backup-vault --backup-vault-name SysOps-Lab06-REPLACE_SUFFIX
aws backup create-backup-plan --backup-plan file://backup-plan.json | tee
evidence/backup-plan.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 4 — Create the resource selection from `backup-selection.json`; target only resources tagged for this course.

```
aws backup create-backup-selection --backup-plan-id REPLACE_PLAN_ID --backup-
selection file://backup-selection.json | tee evidence/selection.json
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 5 — Start an on-demand backup and record start/completion times, recovery-point ARN and status.

```
aws backup start-backup-job --backup-vault-name SysOps-Lab06-REPLACE_SUFFIX --
resource-arn REPLACE_RESOURCE_ARN --iam-role-arn REPLACE_BACKUP_ROLE_ARN | tee
evidence/backup-job.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 6 — Restore to an isolated learner-owned target; never overwrite the original resource.

```
aws backup start-restore-job --recovery-point-arn REPLACE_RECOVERY_POINT_ARN --iam-
role-arn REPLACE_RESTORE_ROLE_ARN --metadata file://restore-metadata.json | tee
evidence/restore-job.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 7 — Validate restored object count, checksum or application-readable state; record the exact method and result.

```
shasum -a 256 sample-data.txt | tee evidence/source-checksum.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 8 — Complete `RECOVERY-REPORT.md` with actual RTO, observed RPO, integrity result, gaps and owners.

```
open RECOVERY-REPORT.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 9 — Capture backup and restore job status as JPEG evidence and compare with the architecture reference.

```
open EXPECTED-ARCHITECTURE.jpg
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 10 — Run validation; remove the restored target and learner-created plan after retention/permission checks.

```
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Verification

A completed recovery point is restored to an isolated target, integrity is checked, actual recovery is compared with RTO/RPO, gaps have owners, and validation emits LAB 06 PASS.

Evidence to retain

- Backup plan and selection
- Backup job
- Restore job
- checksum evidence
- completed RECOVERY-REPORT.md
- JPEG evidence
- cleanup.txt

Acceptance criteria

- Scope uses course tags
- Restore target is isolated
- Retention is explicit
- Integrity validation is objective
- Actual RTO/RPO and gaps are recorded

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Lab 07 — Build a Multi-Account Compliance Evidence Flow

Field	Lab specification
Day / topic	Day 2 · Topic 07
Objective	Apply K1, K3, A1 and A7 by evaluating required tags and encryption, aggregating findings and documenting a controlled remediation exception.
Scenario	Northstar Retail cannot demonstrate consistent security and service-quality controls across its training accounts.
Deliverable	A Config rule template, tag policy, findings register, remediation decision and service-quality scorecard.
Services	AWS Config · Organizations · Security Hub · EventBridge · Systems Manager Automation
Visual	EXPECTED-ARCHITECTURE.jpg

Before you begin

- Confirm caller identity and Region.
- Read every supplied editable YAML, JSON, shell, Python or CLI asset.
- Resolve every REPLACE placeholder and use a unique learner suffix.
- Define expected service, security, performance and cleanup evidence before changing AWS.

Step-by-step procedure

Step 1 — Review the required state in `QUALITY-STANDARD.md`; define scope, evidence, owner, remediation SLA and exception expiry.

```
open QUALITY-STANDARD.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 2 — Validate `config-controls.yaml` and `tag-policy.json`; replace the approved delivery channel and organisation values.

```
aws cloudformation validate-template --template-body file://config-controls.yaml
python3 -m json.tool tag-policy.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 3 — Deploy the Config controls in the learner account and record the recorder and delivery status.

```
aws cloudformation deploy --stack-name SysOps-Lab07-REPLACE_SUFFIX --template-file
config-controls.yaml --capabilities CAPABILITY_NAMED_IAM
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 4 — Create one compliant and one intentionally noncompliant learner-owned test resource.

```
aws resourcegroupstaggingapi get-resources --tag-filters
Key=Course,Values=TGS-2024051413 | tee evidence/tagged-resources.json
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 5 — Run rule evaluation and export compliant/noncompliant resources with timestamps.

```
aws configservice get-compliance-details-by-config-rule --config-rule-name required-tags | tee evidence/config-findings.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 6 — Review `sample-security-findings.json`; prioritise by severity, exposure, asset and exploitability—not count alone.

```
python3 -m json.tool sample-security-findings.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 7 — Complete `REMEDIATION-REGISTER.csv` with owner, due date, chosen automation and verification.

```
open REMEDIATION-REGISTER.csv
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 8 — Document one time-bound exception with business owner, compensating control and expiry.

```
open EXCEPTION-RECORD.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 9 — Capture Config compliance and finding summary as JPEG evidence; run validation.

```
open EXPECTED-ARCHITECTURE.jpg
bash validate.sh
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 10 — Remove intentional noncompliance and delete learner-created controls; retain the evidence files.

```
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

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Verification

Required controls evaluate learner-owned resources, findings are prioritised with context, remediation has owner/SLA/verification, the exception expires, and validation emits LAB 07 PASS.

Evidence to retain

- Config findings
- tagged resources export
- completed remediation register
- exception record
- JPEG evidence
- cleanup.txt

Acceptance criteria

- Scope and standard are explicit
- Every finding has an owner
- Exception has compensating control and expiry
- Verification proves compliant state
- Intentional noncompliance is removed

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Lab 08 — Diagnose IAM Access and Protect Data with KMS

Field	Lab specification
Day / topic	Day 2 · Topic 07
Objective	Apply K1, K3, A1 and A7 by explaining an access decision, narrowing permissions and verifying encryption and key-policy behaviour.
Scenario	A learner workload is denied access to an encrypted object after a policy change, while an earlier role remains broader than required.
Deliverable	A policy-evaluation worksheet, corrected least-privilege policy, KMS key-policy review and encrypted-object evidence.
Services	IAM · Access Analyzer · Policy Simulator · KMS · S3 · CloudTrail
Visual	EXPECTED-ARCHITECTURE.jpg

Before you begin

- Confirm caller identity and Region.
- Read every supplied editable YAML, JSON, shell, Python or CLI asset.
- Resolve every REPLACE placeholder and use a unique learner suffix.
- Define expected service, security, performance and cleanup evidence before changing AWS.

Step-by-step procedure

Step 1 — Read `ACCESS-SCENARIO.md`; identify principal, action, resource, conditions, boundary and possible explicit denies.

```
open ACCESS-SCENARIO.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 2 — Validate the supplied identity, bucket and KMS policy JSON before reasoning about the decision.

```
for f in identity-policy.json bucket-policy.json kms-key-policy.json; do python3 -m
json.tool "$f" >/dev/null; done
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 3 — Use the IAM policy simulator or trainer-provided equivalent to test GetObject on the exact learner object ARN.

```
aws iam simulate-principal-policy --policy-source-arn REPLACE_ROLE_ARN --action-
names s3:GetObject kms:Decrypt --resource-arns REPLACE_OBJECT_ARN REPLACE_KEY_ARN |
tee evidence/simulation.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 4 — Inspect CloudTrail for the denied request and record error code, principal, resource and UTC time.

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```
aws cloudtrail lookup-events --lookup-attributes
AttributeKey=EventName,AttributeValue=GetObject --max-results 20 | tee
evidence/cloudtrail-deny.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 5 — Complete `ACCESS-DECISION.md` in evaluation order; distinguish the S3 allow from the KMS decrypt requirement.

```
open ACCESS-DECISION.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 6 — Edit `identity-policy.json` to the exact bucket prefix and key; retain no wildcard action.

```
open identity-policy.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 7 — Upload and retrieve a small encrypted learner-owned object, then verify encryption metadata.

```
aws s3api head-object --bucket REPLACE_BUCKET --key lab08/evidence.txt | tee
evidence/head-object.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 8 — Run Access Analyzer validation on the policies and resolve ERROR findings.

```
aws accessanalyzer validate-policy --policy-document file://identity-policy.json --
policy-type IDENTITY_POLICY | tee evidence/policy-validation.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 9 — Capture policy simulation and encrypted-object evidence as JPEG screenshots; run validation.

```
open EXPECTED-ARCHITECTURE.jpg
bash validate.sh
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 10 — Remove the learner test object and any temporary role/policy; do not schedule KMS key deletion without trainer approval.

```
bash cleanup.sh | tee evidence/cleanup.txt
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Verification

The decision path explains the deny, corrected policy uses exact actions/resources, encrypted-object metadata is verified, Access Analyzer reports no ERROR, and validation emits LAB 08 PASS.

Evidence to retain

- Policy simulation
- CloudTrail denied event
- policy validation
- encrypted object metadata
- completed ACCESS-DECISION.md
- JPEG evidence
- cleanup.txt

Acceptance criteria

- No wildcard action
- Resource scope is exact
- KMS and S3 permissions both considered
- Explicit deny precedence explained
- Temporary access is removed

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Lab 09 — Troubleshoot a VPC Connectivity Path

Field	Lab specification
Day / topic	Day 3 · Topic 08
Objective	Apply K2 and A2 by isolating a failed request through DNS, routes, security groups, network ACLs, target health and VPC Flow Logs.
Scenario	A private application instance is healthy locally but cannot reach an approved service endpoint after a network change.
Deliverable	A VPC diagnostic worksheet, reachability/flow-log evidence, corrected network control and before/after result.
Services	VPC · Reachability Analyzer · Flow Logs · Security Groups · NACLs · Endpoints
Visual	EXPECTED-ARCHITECTURE.jpg

Before you begin

- Confirm caller identity and Region.
- Read every supplied editable YAML, JSON, shell, Python or CLI asset.
- Resolve every REPLACE placeholder and use a unique learner suffix.
- Define expected service, security, performance and cleanup evidence before changing AWS.

Step-by-step procedure

Step 1 — Review `PATH-DIAGRAM.md`; write source, destination, protocol, port, expected route and success response.

```
open PATH-DIAGRAM.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 2 — Run `diagnostics.sh` with the learner source and destination; retain every command result even when it fails.

```
bash diagnostics.sh REPLACE_SOURCE_ENI REPLACE_DESTINATION_IP | tee
evidence/diagnostics.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 3 — Confirm DNS resolution and compare the returned address with the intended private/public path.

```
dig +short REPLACE_HOSTNAME | tee evidence/dns.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 4 — Inspect source and destination subnet route tables using longest-prefix-match reasoning.

```
aws ec2 describe-route-tables --filters Name=association.subnet-
id,Values=REPLACE_SUBNET_ID | tee evidence/routes.json
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 5 — Inspect security-group rules in both directions and state whether return traffic is automatically allowed.

```
aws ec2 describe-security-groups --group-ids REPLACE_SG_ID | tee evidence/security-group.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 6 — Inspect network ACL rules by number in both directions, including ephemeral return ports.

```
aws ec2 describe-network-acls --filters Name=association.subnet-id,Values=REPLACE_SUBNET_ID | tee evidence/nacl.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 7 — Create or run Reachability Analyzer where supported and record the first blocking component.

```
aws ec2 start-network-insights-analysis --network-insights-path-id REPLACE_PATH_ID | tee evidence/reachability.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 8 — Query Flow Logs for ACCEPT/REJECT evidence correlated to source, destination, port and UTC time.

```
open FLOW-LOG-QUERY.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 9 — Change only the identified learner-owned control; repeat the same test and record before/after evidence.

```
open DIAGNOSTIC-REPORT.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 10 — Capture route, rule and result as JPEG evidence; run validation and restore temporary changes.

```
open EXPECTED-ARCHITECTURE.jpg
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Verification

The report follows the path in order, names the first failed boundary with evidence, changes one learner-owned control, verifies success with the same test, and validation emits LAB 09 PASS.

Evidence to retain

- DNS result
- route tables
- security-group and NACL exports
- reachability analysis
- Flow Log query
- completed diagnostic report
- JPEG evidence
- cleanup.txt

Acceptance criteria

- Source/destination/protocol/port explicit
- Route and policy evaluated separately
- Stateful/stateless return path explained
- One variable changed
- Before/after test is identical

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Lab 10 — Operate DNS and Content-Delivery Failover

Field	Lab specification
Day / topic	Day 3 · Topic 08
Objective	Apply K2 and A2 by configuring health-aware DNS and CloudFront behaviour, then proving client-visible routing and cache outcomes.
Scenario	Northstar Retail needs a controlled failover path for a public web endpoint and must distinguish DNS, origin and cache effects during an incident.
Deliverable	A Route 53 health-check definition, failover record plan, CloudFront operations checklist and measured failover report.
Services	Route 53 · Health Checks · CloudFront · S3 · CloudWatch
Visual	EXPECTED-ARCHITECTURE.jpg

Before you begin

- Confirm caller identity and Region.
- Read every supplied editable YAML, JSON, shell, Python or CLI asset.
- Resolve every REPLACE placeholder and use a unique learner suffix.
- Define expected service, security, performance and cleanup evidence before changing AWS.

Step-by-step procedure

Step 1 — Review `DNS-FAILOVER-PLAN.md`; record hosted zone, primary/secondary endpoints, TTL, health criteria and rollback authority.

```
open DNS-FAILOVER-PLAN.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 2 — Validate `health-check.json`; replace the trainer-provided test endpoint and unique caller reference.

```
python3 -m json.tool health-check.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 3 — Create the health check and confirm CloudWatch health status before associating any record.

```
aws route53 create-health-check --caller-reference SysOps-Lab10-REPLACE_SUFFIX --
health-check-config file://health-check.json | tee evidence/health-check.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 4 — Review `failover-records.json`; confirm PRIMARY and SECONDARY identifiers, evaluate-target-health setting and TTL.

```
python3 -m json.tool failover-records.json
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 5 — Apply the trainer-approved learner-zone record change and record the change ID.

```
aws route53 change-resource-record-sets --hosted-zone-id REPLACE_ZONE_ID --change-batch file://failover-records.json | tee evidence/record-change.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 6 — Query the name repeatedly from two resolvers; record answers and TTL before failure injection.

```
for n in {1..5}; do date -u +%FT%TZ; dig +noall +answer REPLACE_NAME; sleep 5; done | tee evidence/dns-before.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 7 — Run the bounded primary failure approved by the trainer; measure health-check change and client answer transition.

```
for n in {1..20}; do date -u +%FT%TZ; dig +short REPLACE_NAME; sleep 10; done | tee evidence/dns-failover.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 8 — Inspect CloudFront response headers for cache status, age and request ID; distinguish cached content from origin recovery.

```
curl -I https://REPLACE_DISTRIBUTION/ | tee evidence/cloudfront-headers.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 9 — Complete `FAILOVER-REPORT.md` with observed TTL, health transition, client transition and limitation.

```
open FAILOVER-REPORT.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 10 — Capture DNS health and CloudFront evidence as JPEGs; run validation and restore the baseline.

```
open EXPECTED-ARCHITECTURE.jpg
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Verification

Health and record evidence are linked, failover timing is measured rather than assumed, cache and DNS effects are distinguished, baseline is restored, and validation emits LAB 10 PASS.

Evidence to retain

- Health check
- record change
- DNS before/failover results
- CloudFront headers
- completed failover report
- JPEG evidence
- cleanup.txt

Acceptance criteria

- Uses learner-owned hosted zone
- Primary and secondary identifiers are explicit
- TTL is measured
- Cache response is distinguished from origin
- Rollback restores baseline

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Lab 11 — Diagnose Storage and Database Performance, Then Recover

Field	Lab specification
Day / topic	Day 3 · Topic 09
Objective	Apply K1, K4, A1, A4 and A6 by correlating EBS/RDS signals, selecting a safe change and validating data recovery.
Scenario	A transactional workload shows rising latency after a change, and the operator must determine whether compute, storage, connections or query behaviour is the bottleneck.
Deliverable	A metric analysis, bottleneck decision, change plan, restore checklist and database recovery evidence.
Services	CloudWatch · EBS · RDS · Performance Insights · AWS Backup
Visual	EXPECTED-ARCHITECTURE.jpg

Before you begin

- Confirm caller identity and Region.
- Read every supplied editable YAML, JSON, shell, Python or CLI asset.
- Resolve every REPLACE placeholder and use a unique learner suffix.
- Define expected service, security, performance and cleanup evidence before changing AWS.

Step-by-step procedure

Step 1 — Open `data/metrics.csv`; define units, timestamps and expected baseline before calculating any conclusion.

```
column -s, -t data/metrics.csv
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 2 — Run `analyse_metrics.py`; inspect the output ranking and confirm the highest deviation is reproducible.

```
python3 analyse_metrics.py | tee evidence/metric-analysis.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 3 — Compare CPU, EBS queue, IOPS, throughput, connections and latency in `BOTTLENECK-DECISION.md`.

```
open BOTTLENECK-DECISION.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 4 — Inspect the learner RDS configuration and CloudWatch metrics for the same interval where available.

```
aws rds describe-db-instances --db-instance-identifier REPLACE_DB --output json |
tee evidence/rds-config.json
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 5 — Review Performance Insights or database-engine evidence for waits, slow statements or locks; record limitations if unavailable.

```
open QUERY-EVIDENCE.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 6 — Propose one reversible action with risk, owner, expected signal and rollback; do not change multiple variables.

```
open CHANGE-PLAN.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 7 — Create a snapshot or approved recovery point and record its timestamp and status.

```
aws rds create-db-snapshot --db-instance-identifier REPLACE_DB --db-snapshot-
identifier sysops-lab11-REPLACE_SUFFIX | tee evidence/snapshot.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 8 — Restore to an isolated learner identifier and verify engine status and a trainer-provided data check.

```
aws rds restore-db-instance-from-db-snapshot --db-instance-identifier sysops-lab11-
restore-REPLACE_SUFFIX --db-snapshot-identifier sysops-lab11-REPLACE_SUFFIX | tee
evidence/restore.json
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 9 — Capture metrics, snapshot and restore evidence as JPEGs; record actual recovery timing.

```
open EXPECTED-ARCHITECTURE.jpg
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 10 — Run validation; delete the isolated restore and learner snapshot when approved.

```
bash validate.sh
bash cleanup.sh | tee evidence/cleanup.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

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Verification

Metric calculations are reproducible, the chosen bottleneck cites evidence, one reversible action is proposed, recovery is isolated and timed, and validation emits LAB 11 PASS.

Evidence to retain

- Metric analysis
- RDS configuration
- query evidence
- snapshot/restore records
- completed bottleneck and change decisions
- JPEG evidence
- cleanup.txt

Acceptance criteria

- Units and baseline stated
- Cause remains a hypothesis until tested
- One variable changed
- Restore does not overwrite source
- Data validation and actual timing recorded

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Lab 12 — Capstone — Resolve a Multi-Service CloudOps Incident

Field	Lab specification
Day / topic	Day 3 · Topic 10
Objective	Apply A1–A7 in one connected incident by detecting impact, isolating the failed boundary, restoring service, validating data and documenting service-quality improvements.
Scenario	After a routine change, Northstar Retail experiences elevated web latency, intermittent 5xx responses, rejected VPC traffic, an incomplete backup job and a security finding.
Deliverable	A timeline, evidence matrix, prioritised hypotheses, recovery runbook, validation record and post-incident action register.
Services	CloudWatch · CloudTrail · VPC Flow Logs · ELB · AWS Backup · Security Hub · Systems Manager
Visual	EXPECTED-ARCHITECTURE.jpg

Before you begin

- Confirm caller identity and Region.
- Read every supplied editable YAML, JSON, shell, Python or CLI asset.
- Resolve every REPLACE placeholder and use a unique learner suffix.
- Define expected service, security, performance and cleanup evidence before changing AWS.

Step-by-step procedure

Step 1 — Read every file in `data/`; record evidence source, timestamp, timezone and reliability in `EVIDENCE-MATRIX.csv`.

```
for f in data/*; do echo "--- $f"; sed -n '1,80p' "$f"; done
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 2 — Run `build_timeline.py`; verify the merged timeline is chronologically sorted and retains source IDs.

```
python3 build_timeline.py | tee evidence/timeline-build.txt
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 3 — Frame impact, scope and declared severity without naming a root cause yet.

```
open INCIDENT-RECORD.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 4 — Rank at least three hypotheses and name one discriminating test for each.

```
open HYPOTHESES.csv
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

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Step 5 — Identify the first failed boundary by correlating ALB health, Flow Logs, change events and application latency.

```
open output/timeline.csv
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 6 — Review backup and security evidence; decide which items block recovery and which become follow-up actions.

```
open EVIDENCE-MATRIX.csv
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 7 — Complete `RECOVERY-RUNBOOK.md` with authority, exact action, safety check, verification, communication and rollback.

```
open RECOVERY-RUNBOOK.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 8 — Execute the trainer-approved simulation action or tabletop step; record UTC timestamps and observed outputs.

```
open RECOVERY-VALIDATION.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 9 — Validate service health, request success, data integrity, security control and recovery objective.

```
open RECOVERY-VALIDATION.md
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 10 — Complete `ACTIONS.csv` with preventive, detective and recovery controls, owner, due date and verification threshold.

```
open ACTIONS.csv
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 11 — Deliver a five-minute operational briefing: impact, evidence, decision, recovery, limitation and next controls.

```
open INCIDENT-RECORD.md
```


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Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Step 12 — Capture the final dashboard/architecture evidence as JPEG and run the capstone validator.

```
open EXPECTED-ARCHITECTURE.jpg
bash validate.sh
```

Evidence checkpoint: record the observed state, timestamp and resource identifier before continuing.

Verification

All evidence IDs reconcile, the timeline precedes causal claims, recovery and rollback are explicit, validation covers service and data, A1–A7 are mapped, and validation emits LAB 12 PASS.

Evidence to retain

- timeline.csv
- evidence matrix
- hypothesis register
- incident record
- recovery runbook
- recovery validation
- ACTIONS.csv
- five-minute briefing
- JPEG evidence

Acceptance criteria

- A1–A7 traceability complete
- Observation and hypothesis separated
- First failed boundary supported
- Recovery includes rollback and communication
- Every action has owner, due date and measurable verification

Assessment Readiness

Instrument	Count	Duration	Coverage	Practice evidence
WA-SAQ	5 open-ended questions	60 min	K1–K5	Concept explanations across Topics 01–10
Case Study	5 connected tasks	60 min	A1–A7	Labs 01–12, especially integrated Lab 12

Criterion	Assessment focus	Taught in	Practised in
K1	Cloud infrastructure components and technology	Topics 01, 04, 08, 09	Labs 04, 09–11
K2	Infrastructure configuration and troubleshooting	Topics 02, 08, 10	Labs 01–02, 09–10, 12
K3	Supporting package and fleet operations	Topics 03, 06	Labs 03, 05

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K4	Application design, maintenance and troubleshooting	Topics 04–10	Labs 04–12
K5	Virtualisation technologies	Topics 01, 03, 04	Labs 03–04
A3, A5	Availability and controlled release	Topics 04, 06	Labs 04–05
A1	Monitoring, backup and recovery	Topics 02, 05, 09	Labs 01–02, 06, 11
A7	Service quality and compliance	Topic 07	Lab 07
A4, A6	Issue analysis, continuity and recovery	Topics 05, 10	Labs 06, 12
A2	Diagnostics at scale	Topics 08, 10	Labs 09–10, 12

Operational note: Use your own words and justify each recommendation with observed evidence, risk, verification and rollback. The WSQ assessment is separate from the AWS certification examination.

Glossary

Term	Meaning
Alarm	A stateful evaluation of a metric or expression against defined criteria.
Blast radius	The maximum scope of impact from a failure or change.
CloudOps	Disciplined operation of cloud workloads using observable, automated and governed practices.
Drift	Difference between declared desired configuration and actual resource state.
RPO	Maximum acceptable data loss measured in time.
RTO	Maximum acceptable time to restore service.
Runbook	Controlled operational procedure with inputs, authority, verification and recovery.
SLI / SLO	Measured service indicator and the target set for it.
Telemetry	Metrics, logs, traces, events and configuration evidence used to understand a system.

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